

Mitigating Structural Overfitting: A Distribution-Aware Rectification Framework for Missing Feature Imputation

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Abstract

Incomplete node features are ubiquitous in real-world scenarios such as user profiling and cold-start recommendation, which severely hinders the practical deployment of graph learning systems (e.g., GNNs). Existing solutions typically rely on diffusion-based structural smoothing (e.g., feature propagation) to impute missing values. However, we find that these approaches suffer from structural overfitting, leading to three progressive challenges: 1) performance degradation on disjoint graphs, 2) loss of semantic diversity due to over-smoothing, and 3) feature distribution shift when generalizing to unseen graph structures (inductive tasks). To address these challenges, we introduce the **DART** framework. It begins by employing *Global Structural Augmentation (GSA)*, which establishes global correlations to bridge disjoint components and extend diffusion coverage. Building upon this, we design a semantic rectifier based on masked autoencoding. This module learns the latent feature manifold to recover natural semantic details. Crucially, we introduce a test-time distribution rectification mechanism that projects structurally biased features back onto the learned manifold during inference, effectively bridging the inductive distribution gap. Furthermore, considering that synthetic masking fails to reflect real-world sparsity, we present a new dataset **Sailing** collected from voyage records with naturally missing attributes. Extensive experiments on six public datasets and Sailing demonstrate that DART significantly outperforms state-of-the-art methods in both transductive and inductive settings. Our code and dataset are available at <https://github.com/yfsong00/DART>.

*These authors contributed equally to this research.

Keywords

Incomplete Graphs, Test-time Adaptation, Feature Imputation

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1 Introduction

Graphs are important structures in the real world, which are capable of modeling complex interactions among various objects. Graph learning approaches such as graph neural networks (GNNs) [3, 6, 10, 14, 18, 20, 33] have demonstrated impressive effectiveness in a wide range of applications for graph analysis, including trajectory prediction [23, 25] and user classification [20, 35–37]. These methods typically achieve robust embeddings by combining graph feature information and graph structure. However, a critical bottleneck in real-world deployments is the ubiquity of *incomplete node features*. A prominent example arises in the domain of maritime navigation, where critical features such as vessel velocity and wind conditions may be incomplete or missing due to signal interference or sensor limitations, seriously degrading the performance of graph learning methods [27, 31].

To address this issue, recent studies have proposed various imputation techniques [15, 17, 30]. Among them, diffusion-based methods based on feature propagation (FP) [27] have achieved superior performance by iteratively diffusing known features over the graph topology to reconstruct missing features. However, we argue that they fundamentally suffer from *structural overfitting*, where imputed values are deterministically conditioned on graph topology while ignoring the intrinsic feature manifold. This dependency introduces three critical challenges: 1) The imputation quality is highly sensitive to graph connectivity. As illustrated in Figure 1a, previous studies only validate their effectiveness on the largest connected component, which is unreasonable. The performance of diffusion-based methods (FP and PCFI) is significantly degraded on whole graphs with multiple connected components. 2) These methods face over-smoothing problem in transductive tasks. Figure

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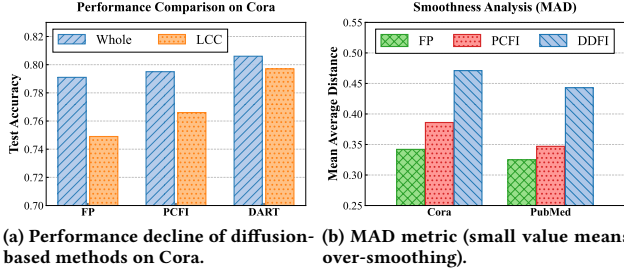


Figure 1: Analysis of Structural Overfitting in FP-based methods. LCC means validate on the largest connected component (previous work’s unreasonable setting) while Whole means validate on whole graph (our setting).

1b reports the MAD metric [4], the small values of diffusion-based methods indicate a severe reduction in feature diversity. 3) Structural overfitting leads a feature distribution shift between training and inference in inductive tasks. As shown in Figure 2, the structure of the partial training graph differs from the complete inference graph. It causes the imputed feature distributions to diverge, thereby affecting the generalization performance in downstream tasks.

To tackle these challenges, we propose **DART** (Distribution-Aware Rectification via Test-time adaptation), a framework that decouples structural estimation from semantic manifold learning. First, we design **Global Structural Augmentation (GSA)** that utilizes label consistency to establish semantic bridges between disjoint components, which extends the diffusion scope. Second, to restore semantic diversity, we train a masked autoencoder as the manifold learner to reconstruct features from structurally perturbed inputs. Crucially, a Gaussian-based masking strategy is used to force the model to capture the robust feature manifold rather than learning smoothed values. Finally, to bridge the inductive gap, we introduce a test-time distribution rectification mechanism. During inference, we project the structurally biased backs onto the learned manifold via the encoder-decoder framework, effectively correcting the distribution shift.

Furthermore, existing methods typically evaluate performance by randomly masking features in complete datasets (synthetic missing), which fails to reflect the complexity of real-world sparsity. To benchmark imputation methods under realistic scenes, we present a new dataset called **Sailing** collected from real maritime voyage records provided by the Danish Maritime Authority (DMA) [2]. Notably, 80.4% of the features in Sailing are *naturally missing*, providing a rigorous dataset for real-world deployment.

In summary, our main contributions are:

- We identify the structural overfitting issue in existing diffusion-based methods, which leads to limited performance on disjoint graphs, low diversity and distribution shift.
- We propose DART, a novel framework that integrates a manifold learner with test-time distribution rectification, effectively improving feature diversity and mitigating inductive distribution shift.
- We release Sailing, a real-world maritime dataset with naturally missing attributes. Extensive experiments on Sailing

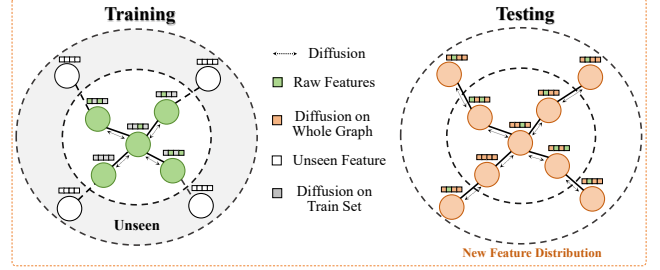


Figure 2: Illustration of Inductive Distribution Shift caused by structural changes.

and six public datasets demonstrate that DART significantly outperforms state-of-the-art methods.

2 Preliminaries

2.1 Notations and Problem Formulation

Let $G = (\mathcal{V}, \mathcal{E})$ be an undirected graph with $|\mathcal{V}| = N$ nodes. We denote the adjacency matrix as $\mathbf{A} \in \mathbb{R}^{N \times N}$ and the node label vector as $\mathbf{Y} \in \mathbb{R}^N$. The node feature matrix is denoted as $\mathbf{X} \in \mathbb{R}^{N \times d}$. Consider a realistic scenario where node features are partially observed. Let $\mathbf{M} \in \{0, 1\}^{N \times d}$ be a binary mask matrix, where $M_{ij} = 1$ indicates the j -th feature of node v_i is observed, and 0 otherwise. The observed feature matrix is given by $\mathbf{X}_{obs} = \mathbf{X} \odot \mathbf{M}$.

Problem Formulation. The goal of missing feature imputation is to learn a mapping function $\Phi : (\mathbf{X}_{obs}, \mathbf{A}) \rightarrow \hat{\mathbf{X}}$ that reconstructs the complete feature matrix $\hat{\mathbf{X}}$. The imputed features are then used by a downstream model f_θ (e.g., a GNN) to minimize the task-specific loss (e.g., cross-entropy for node classification):

$$\min_{\Phi, \theta} \mathcal{L}_{task}(f_\theta(\Phi(\mathbf{X}_{obs}, \mathbf{A}), \mathbf{A}), \mathbf{Y}). \quad (1)$$

We consider both *transductive* settings (where $\mathbf{A}_{test} = \mathbf{A}_{train}$) and *inductive* settings (where $\mathbf{A}_{test}$ contains unseen nodes and edges disjoint from $\mathbf{A}_{train}$).

Inductive Feature Distribution Shift. In inductive graph learning, the structural contexts differ significantly between training and testing. We denote the *feature distribution shift* Δ_S in inductive tasks as the discrepancy between the distributions of features imputed strictly via structural propagation on disjoint topologies. Formally, let $P_\Phi(\hat{\mathbf{X}}|\mathbf{A})$ denote the conditional distribution of features reconstructed by a structure-dependent mapping Φ (e.g., FP). The shift is defined as:

$$\Delta_S = \mathcal{F}\left(P_\Phi(\hat{\mathbf{X}}|\mathbf{A}_{test}) \parallel P_\Phi(\hat{\mathbf{X}}|\mathbf{A}_{train})\right), \quad (2)$$

where $\mathcal{F}$ is a divergence measure (e.g., Kullback-Leibler divergence). Crucially, since P_Φ is conditioned on the graph topology, structural differences between $\mathbf{A}_{train}$ and $\mathbf{A}_{test}$ introduce a structural bias δ_{struct} , causing the imputed support to deviate from the true manifold $\mathcal{M}$ (i.e., introducing "off-manifold" noise), which serves as the primary challenge we aim to rectify.

2.2 Revisiting Feature Propagation

Feature Propagation (FP) [27] serves as the foundational diffusion mechanism for imputing the missing node features. It assumes that valid feature signals should vary smoothly across the topological structure. Here we formulate FP as a discrete Dirichlet boundary value problem on the graph. Specifically, the reconstruction of missing features aims to minimize the graph Dirichlet energy while strictly adhering to the observed signals:

$$\mathcal{E}(\tilde{\mathbf{X}}) = \text{tr}(\tilde{\mathbf{X}}^\top \tilde{\mathbf{L}} \tilde{\mathbf{X}}), \quad \text{s.t. } \tilde{\mathbf{X}}_{\mathcal{V}_{obs}} = \mathbf{X}_{obs}, \quad (3)$$

where $\tilde{\mathbf{L}} = \mathbf{I} - \tilde{\mathbf{A}}$ denotes the normalized graph Laplacian, and the constraint $\tilde{\mathbf{X}}_{\mathcal{V}_{obs}} = \mathbf{X}_{obs}$ ensures fidelity to the ground truth. The optimal solution corresponds to the steady state of a heat diffusion process constrained by boundary conditions. In practice, this harmonic solution is efficiently approximated via an iterative diffusion-and-reset procedure:

$$\mathbf{X}^{(k+1)} = (\mathbf{I} - \mathbf{M}) \odot (\tilde{\mathbf{A}}\mathbf{X}^{(k)}) + \mathbf{M} \odot \mathbf{X}_{init}, \quad (4)$$

where $\mathbf{X}^{(0)} = \mathbf{X}_{init}$ (typically zero-filled). In each iteration, the feature information propagates from neighbors through the normalized adjacency $\tilde{\mathbf{A}}$, effectively acting as a low-pass filter that smooths high-frequency signals. Meanwhile, the term $\mathbf{M} \odot \mathbf{X}_{init}$ enforces the boundary condition by explicitly resetting the observed nodes to their original values. This process repeats until convergence, yielding the structurally smoothed estimation $\tilde{\mathbf{X}}_{fp}$. These issues indicate that FP-based imputations are inherently biased towards the current graph topology and ignore the latent manifold of the real features, which motivates our proposed DART framework.

3 The DART Framework

To tackle the above challenges of previous feature imputation methods, we propose DART, a framework combines global structural augmentation with a masked autoencoder to recover semantic manifolds and addresses the distribution shift in graph feature imputation through a novel test-time adaptation strategy. Unlike previous methods that strictly rely on structural diffusion, DART functions as a decoupling framework that separates *structural estimation* from *semantic manifold learning*. As illustrated in Figure 3, the overall workflow consists of three progressive stages.

First, **Global Structural Augmentation (GSA)** constructs a global affinity topology to bridge disconnected components, ensuring valid structural initialization. Second, a **Self-Supervised Manifold Learner** utilizes a graph-based masked autoencoder with continuous semantic perturbation to capture the intrinsic feature distribution, thereby recovering semantic diversity. Finally, to bridge the generalization gap in inductive tasks, we introduce a **Test-Time Distribution Rectification** mechanism, which projects structurally biased estimates back onto the learned manifold during inference to align the feature distribution.

3.1 Stage I: Global Structural Augmentation

As analyzed in Section 2, diffusion-based methods rely on valid propagation paths to impute missing features. We formalize the relationship between topological connectivity and imputation effectiveness with the following lemma:

Algorithm 1: Global Structural Augmentation (GSA)

Input: Training Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, Features $\mathbf{X}$, Labels $\mathbf{Y}_{train}$, Parameters k, τ, δ .

Output: Augmented Graph $\tilde{\mathcal{G}} = (\mathcal{V}, \tilde{\mathcal{E}})$.

$\tilde{\mathcal{E}} \leftarrow \mathcal{E}$;

foreach node $v_i \in \mathcal{V}$ **do**

 // 1. Similarity Computation

 Compute weights: $w_j \leftarrow \text{sim}(\mathbf{x}_i, \mathbf{x}_j), \forall v_j \in \mathcal{V}$;

 Normalize probability: $p_j \leftarrow \frac{\exp(w_j/\tau)}{\sum_{v_l \in \mathcal{V}} \exp(w_l/\tau)}$;

 // 2. Semantically-Guided Linking

 Sample set $\mathcal{S}_i$ of size k from $\mathcal{V}$ with distribution $\{p_j\}$;

foreach node $v_j \in \mathcal{S}_i$ **do**

 // Filter by label consistency

$\mathcal{V}_{link} \leftarrow \{v_j \in \mathcal{S}_i \mid y_j = y_i\}$;

 // Filter by similarity threshold

$\mathcal{V}_{link} \leftarrow \{v_j \in \mathcal{S}_i \mid w_j > \delta \cap v_j \notin \mathcal{V}_{link}\}$;

$\tilde{\mathcal{E}} \leftarrow \tilde{\mathcal{E}} \cup (\{v_i\} \times \mathcal{V}_{link})$;

return $(\mathcal{V}, \tilde{\mathcal{E}})$;

LEMMA 3.1. (*Diffusion Reachability Bound*). Let $\mathbf{S} = \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$ be the normalized adjacency matrix. The imputation of a missing node v_i via feature propagation is theoretically given by a weighted sum of observed features over all possible paths. If v_i belongs to a connected component with no observed nodes (i.e., disjoint from $\mathcal{V}_{obs}$), the imputed value is strictly zero, leading to maximum error.

PROOF. Denote $\mathcal{O}$ and $\mathcal{U}$ as the sets of observed and missing nodes respectively, we first partition the matrix $\mathbf{S}$ and feature vector $\mathbf{X}$ according to $\mathcal{O}$ and $\mathcal{U}$:

$$\mathbf{S} = \begin{bmatrix} \mathbf{S}_{\mathcal{O}\mathcal{O}} & \mathbf{S}_{\mathcal{O}\mathcal{U}} \\ \mathbf{S}_{\mathcal{U}\mathcal{O}} & \mathbf{S}_{\mathcal{U}\mathcal{U}} \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} \mathbf{X}_{\mathcal{O}} \\ \mathbf{X}_{\mathcal{U}} \end{bmatrix}. \quad (5)$$

The optimal solution of feature propagation for the missing nodes $\mathbf{X}_{\mathcal{U}}$ is obtained by setting the partial derivative of the Dirichlet energy with respect to $\mathbf{X}_{\mathcal{U}}$ to zero, yielding the closed-form harmonic solution:

$$\mathbf{X}_{\mathcal{U}} = (\mathbf{I} - \mathbf{S}_{\mathcal{U}\mathcal{U}})^{-1} \mathbf{S}_{\mathcal{U}\mathcal{O}} \mathbf{X}_{\mathcal{O}}. \quad (6)$$

Since the spectral radius $\rho(\mathbf{S}_{\mathcal{U}\mathcal{U}}) < 1$, we can expand the inverse term using the Neumann Series. For a specific missing node $v_i \in \mathcal{U}$, its imputed value is:

$$\hat{\mathbf{x}}_i = \sum_{v_j \in \mathcal{O}} \left(\sum_{k=0}^{\infty} \sum_{v_m \in \mathcal{U}} (\mathbf{S}_{\mathcal{U}\mathcal{U}}^k)_{im} (\mathbf{S}_{\mathcal{U}\mathcal{O}})_{mj} \right) \mathbf{x}_j. \quad (7)$$

This expansion has a clear topological interpretation: the term $(\mathbf{S}_{\mathcal{U}\mathcal{U}}^k)_{im} (\mathbf{S}_{\mathcal{U}\mathcal{O}})_{mj}$ represents a path of length $k+1$ starting from an observed node v_j , passing through a neighbor $v_m \in \mathcal{U}$, and traversing k hops within the missing set $\mathcal{U}$ to reach v_i . If v_i resides in a component disjoint from $\mathcal{O}$, there exists no path connecting any observed node v_j to v_i , which leads to $\hat{\mathbf{x}}_i = \mathbf{0}$. Assuming the ground truth $\mathbf{x}_i \neq \mathbf{0}$, the imputation error is maximized. Thus, structural reachability is a prerequisite for valid diffusion-based imputation. $\square$

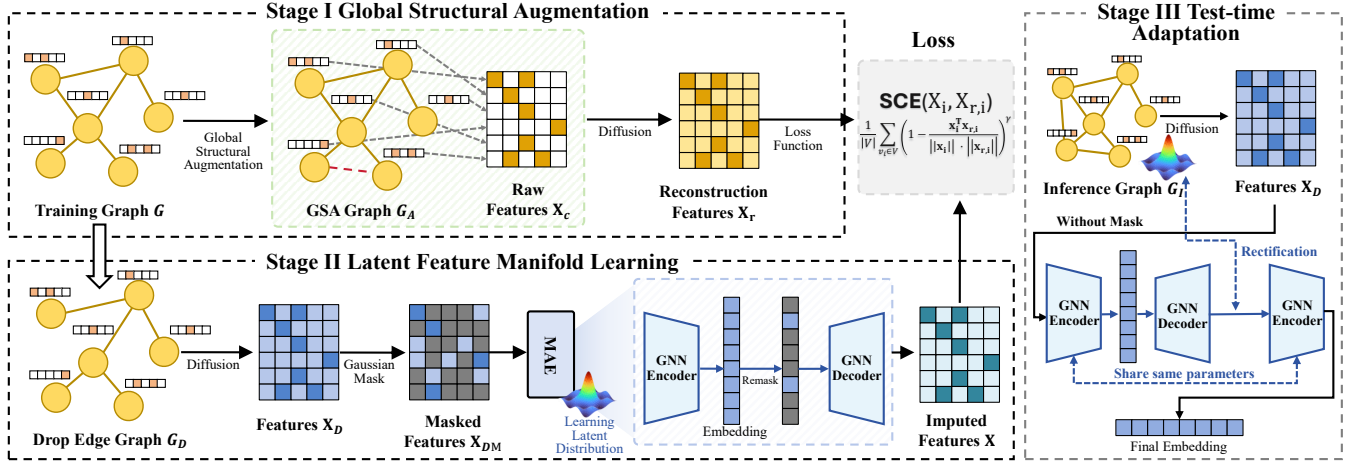


Figure 3: Training and inference process of DART.

Lemma 3.1 indicates that optimizing the diffusion equation is invalid for topologically isolated components. To mitigate this issue, a naive strategy is to randomly link node pairs across different components. However, this approach ignores the semantic consistency of nodes, inevitably introducing noise that damages the quality of imputed features [27].

To alleviate the unreachability problem while maintaining high reliability, we propose **Global Structural Augmentation (GSA)**. Instead of random linking, GSA leverages available supervision signals and high-confidence feature similarities to establish semantic bridges. Specifically, as outlined in Algorithm 1, for labeled nodes, we strictly filter global neighbors based on label consistency. For unlabeled nodes, we only retain links with feature similarity exceeding a threshold δ . By constructing the augmented graph $\tilde{G}$, GSA introduces long-range edges that connect potentially disjoint components. Through this selective linking strategy, GSA provides a structurally enhanced initialization that permits information flow across previously disconnected regions while minimizing the introduction of noisy edges.

3.2 Stage II: Self-Supervised Manifold Learning

After GSA, we obtain an augmented graph $\tilde{G}$. While it establishes long-range semantic bridges to alleviate topological isolation, the features initialized by diffusion on $\tilde{G}$ still suffer from over-smoothing problem. To recover the latent semantic patterns (i.e. the distribution of the real features), we employ a graph-based Masked Autoencoder (MAE) framework to learn the structurally decoupled feature distribution with the following strategy.

3.2.1 Dual-Perturbation Strategy. Standard MAEs typically use binary masking on raw features. However, we argue that this setting is insufficient for missing feature imputation for two reasons. First, the inputs are already smoothed results from propagation, making binary masking a trivial reconstruction task that fails to capture complex dependencies. Second, standard training assumes a static graph structure, leaving the model vulnerable to the structural

shifts inherent in inductive tasks. To address these issues, DART incorporates a dual-perturbation strategy to perceive the distribution of real features that are not strongly correlated with the structure.

Structural Perturbation for Inductive Robustness. In inductive settings, the local topology of a node v_i in the inference graph A_{test} often differs from that in A_{train} . If the model over-relies on the specific edges in A_{tr} , it will fail to generalize. To force the model to learn representations invariant to local topological changes, we randomly drop edges from the training graph to generate a perturbed view A_{drop} in each epoch. The input features are then generated by propagating raw features on this partial structure:

$$X_{in} = \tilde{A}_{drop} X_{in} \odot (1 - M) + X_{tr} \odot M, \quad (8)$$

where $\tilde{A}_{drop}$ is the normalized adjacency matrix, and M is the feature-level binary mask. This equation represents the steady state where information has been fully propagated across the perturbed graph while preserving the observed values. By training on incomplete structures A_{drop} , it forces the encoder to reduce its dependency on specific edges and focus more on the underlying feature distribution.

Continuous Semantic Perturbation. Instead of binary masking, we apply a Gaussian-based continuous masking strategy. Since the FP-imputed inputs X_{in} are dense continuous values, binary masking (setting to zero) creates an artificial distribution gap. In contrast, adding continuous noise simulates the natural uncertainty of the manifold. We generate the perturbed input $\tilde{X}$ as:

$$W \sim \mathcal{N}(0, 1), \quad \tilde{X} = X_{in} \odot \sigma(W + \delta), \quad (9)$$

where σ is the sigmoid function and δ controls the noise level. According to our theoretical analysis in Appendix 3.4, reconstructing representation from continuous noise forces the model to learn the vector field pointing towards the high-density regions of the data distribution. This effectively prevents the model from overfitting of the smoothed values and enables the recovery of high-frequency discriminative details.

3.2.2 Manifold Learning Objective. To guide the model learn the correct semantic manifold, we set the features propagated on the

GSA graph $\tilde{\mathcal{G}}$ as the reconstruction target, denoted as $\mathbf{X}_{target} = \text{FP}(\mathbf{X}_{tr}, \tilde{\mathcal{G}})$. Let $\hat{\mathbf{X}} = \text{Decoder}(\text{Encoder}(\tilde{\mathbf{X}}))$ be the reconstructed output from the perturbed input. We optimize the model using the Scaled Cosine Error (SCE) loss:

$$\mathcal{L} = \frac{1}{|\mathcal{V}|} \sum_{v_i \in \mathcal{V}} \left(1 - \frac{\hat{\mathbf{x}}_i^\top \mathbf{x}_{target,i}}{\|\hat{\mathbf{x}}_i\| \cdot \|\mathbf{x}_{target,i}\|} \right)^\gamma, \quad (10)$$

where $\gamma \geq 1$ is a focusing parameter. By mapping the input from a structurally unstable view ($\mathbf{A}_{drop}$) to a structurally augmented target ($\tilde{\mathcal{G}}$), the model is forced to ignore noise in local structure and capture invariant global semantic patterns (i.e., real-feature distribution and structural invariance).

3.3 Stage III: Test-Time Distribution Rectification

Standard frameworks typically use the trained encoder to generate embeddings directly. However, as defined in Eq. (2), the structural difference in inductive tasks introduces a distribution shift, meaning the FP output on the test graph deviates from the training manifold.

To address this, we propose the test-time rectification strategy. We interpret the trained autoencoder not just as a feature extractor, but as a manifold projector. The inference proceeds in two steps:

Step 1: Biased Estimation. We first perform feature propagation on the test graph to obtain an initial estimation. Due to the structural shift, this estimation is biased: $\hat{\mathbf{X}}_{bias} = \mathbf{A}_{test} \mathbf{X}_{in} \odot (\mathbf{I} - \mathbf{M}) + \mathbf{X}_{tr} \odot \mathbf{M}$.

Step 2: Manifold Projection. We feed $\hat{\mathbf{X}}_{bias}$ into the full encoder-decoder network. Since the model has been trained (via structural perturbation) to reconstruct valid semantics from corrupted structures, it effectively filters out the structural noise in $\hat{\mathbf{X}}_{bias}$:

$$\mathbf{X}_{rec} = \text{Decoder}(\text{Encoder}(\hat{\mathbf{X}}_{bias})). \quad (11)$$

Finally, the rectified features $\mathbf{X}_{rec}$ are passed through the encoder to generate the embeddings $\mathbf{Z} = \text{Encoder}(\mathbf{X}_{rec})$. This re-projection ensures that the input to the downstream classifier lies on the valid feature manifold learned during training, significantly bridging the inductive gap.

3.4 Theoretical Analysis of the DART Framework

In this section, we provide a rigorous theoretical analysis of DART, grounding its effectiveness in the *Manifold Hypothesis*. We demonstrate that the two core components of DART—the self-supervised learner and the test-time rectification—mathematically correspond to score matching and manifold projection, respectively.

Score Matching for Diversity Recovery. The over-smoothing issue in GNNs can be viewed as the collapse of the feature distribution $p_{data}(\mathbf{x})$ towards a low-frequency subspace. We show that the Masked Autoencoder (MAE) in DART counteracts this by learning the score function of the data distribution. Formally, let the corruption process (Gaussian masking) be modeled as $\tilde{\mathbf{x}} = \mathbf{x} + \epsilon$, where $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ is the injected noise vector. The MAE training objective minimizes the expected denoising error:

$$\mathcal{L}_{MAE} = \mathbb{E}_{\mathbf{x} \sim p_{data}} \mathbb{E}_{\epsilon} [\|\mathbf{x} - r_\theta(\tilde{\mathbf{x}})\|_2^2], \quad (12)$$

Table 1: Dataset statistics (D: Graph Density, C: Clustering coefficient).

Dataset	Nodes	Edges	D (10^{-3})	C
Cora	2,708	10,566	1.48	0.261
Citeseer	3,327	9,228	0.85	0.141
PubMed	19,717	88,651	0.22	0.060
Ogbn-Arxiv	169,343	1,166,243	0.08	0.118
Flickr	89,250	899,756	0.22	0.033
Reddit	232,965	23,213,838	0.42	0.240
Sailing	5,014	55,458	2.21	0.323

where $r_\theta(\cdot)$ denotes the encoder-decoder function parameterized by θ . Following the theory of Denoising Autoencoders (DAE) [1], the optimal minimizer $r^*(\cdot)$ asymptotically approximates the gradient of the log-density:

$$r^*(\mathbf{x}) = \mathbf{x} + \sigma^2 \nabla_{\mathbf{x}} \log p_{data}(\mathbf{x}) + o(\sigma^2), \quad (13)$$

where $o(\sigma^2)$ represents the higher-order term that approaches zero faster than σ^2 , the gradient term $\nabla_{\mathbf{x}} \log p_{data}(\mathbf{x})$ defines a vector field pointing towards the high-density regions of the manifold [16]. This derivation implies that DART learns a vector field pointing towards the high-density modes of the manifold $\mathcal{M}$. While structural propagation acts as a low-pass filter that smooths the semantics, Eq. (13) shows that the learned MAE model could effectively push representations back towards high-probability modes, thereby recovering semantic diversity.

Rectification as Manifold Projection. In inductive settings, the structural change introduces a bias into the FP-imputed features. We model the biased estimation $\hat{\mathbf{x}}_{fp}$ as a perturbation of the ground truth $\mathbf{x}_{gt} \in \mathcal{M}$:

$$\hat{\mathbf{x}}_{fp} = \mathbf{x}_{gt} + \delta_{\perp}, \quad (14)$$

where $\delta_{\perp}$ represents the structural noise component orthogonal to the manifold tangent space, which causes the distribution shift. During inference, DART applies the learned function $r^*(\cdot)$ to this biased input. Substituting $\hat{\mathbf{x}}_{fp}$ into Eq. (13), the rectification process yields:

$$\hat{\mathbf{x}}_{rec} = r^*(\hat{\mathbf{x}}_{fp}) \approx \hat{\mathbf{x}}_{fp} + \sigma^2 \nabla_{\mathbf{x}} \log p_{data}(\hat{\mathbf{x}}_{fp}). \quad (15)$$

Geometrically, the term $\nabla_{\mathbf{x}} \log p_{data}(\hat{\mathbf{x}}_{fp})$ represents the direction of steepest ascent on the probability landscape. Since $\hat{\mathbf{x}}_{fp}$ lies in a low-density region due to the shift, this gradient term points directly towards the nearest high-density region on $\mathcal{M}$. Consequently, the two-step inference approximates a gradient ascent step that effectively acts as a projection operator:

$$\hat{\mathbf{x}}_{rec} \approx \hat{\mathbf{x}}_{fp} - \mathcal{P}_{\perp}(\delta_{\perp}) \longrightarrow \mathcal{P}_{\mathcal{M}}(\hat{\mathbf{x}}_{fp}), \quad (16)$$

where $\mathcal{P}_{\mathcal{M}}$ denotes the projection onto the intrinsic manifold. Thus, DART mathematically functions as a manifold projector, filtering out the off-manifold structural noise $\delta_{\perp}$ induced by the inductive feature distribution gap.

4 Experiments

In this section, we evaluate DART compared with eight baselines across two tasks: node classification and link prediction. For node classification, we conduct experiments under both transductive and inductive settings. For link prediction, we evaluate all the methods

as specifically demonstrated in Section 4.4. All experiments are conducted on a single machine with an Intel Xeon 8377C CPU, an NVIDIA RTX 3090 GPU (24GB), and 1TB of RAM.

4.1 Datasets and Baselines

To verify the effectiveness of DART in real-world scenarios, we propose a new dataset for missing feature imputation called **Sailing**, which is collected from Danish Nautical Data Records [2]. We select a subset from the original data and construct the graph by dividing the grid according to latitude and longitude. Trajectory points in the same grid or adjacent grids are connected as edges. Appendix A.2 reports more construction details. After the construction, the Sailing dataset naturally has 80.4% missing features, providing a new benchmark dataset for missing feature imputation methods to validate their effectiveness in real-world scenarios. Next, we evaluate the transductive node classification task on five datasets: Cora, Citeseer, PubMed [28], Ogbn-Arxiv [13] and Sailing. To validate our effectiveness on inductive tasks, we evaluate the inductive node classification task on three datasets: Reddit, Flickr [41] and Sailing. The dataset statistics are shown in Table 1.

DART is compared with eight baselines, including: (1) two standard graph embedding methods: GCN [22] and GAT [33]; and (2) six methods designed for missing feature scenarios: GCNMF [30], PaGNN [17], FP [27], PCFI [31], ASD-VAE [19] and DRI [24]. Following previous work [27, 31], for public datasets, we generate the missing features with *Uniform* and *Structural* missing types as input for all methods. Specifically, uniform missing means that each feature dimension of each node is missing randomly, while structural missing represents that either we observe all features for a node, or we observe none. For the Sailing dataset, we utilize its naturally missing features. For FP and PCFI, we use the same type of GNN as the encoder in DART for a fair comparison.

4.2 Experimental Setup

For the transductive and inductive node classification task, we report the mean and standard error of the test accuracy with running five times. For the link prediction task, AUC and AP scores [7] are used to validate the effectiveness of all the methods following prior work [29, 38, 39]. All results are under 90% missing feature rate (except for Sailing, which naturally has an 80.4% missing feature rate). For DART, we first simply set the $\tau = 1e9$ (uniform sampling) for GSA and follow previous work [12] to set the parameters for MAE to avoid excessive parameter tuning. Then we adjust GSA parameters k and δ to achieve the best result on the validation set, with all details provided in Section 4.6. For other baselines, we use their official code to reproduce the results. Note that distinct from prior studies [9, 19, 31] that often limit evaluation to the Largest Connected Component (LCC), we use the general GNN evaluation protocol that all experiments are conducted on the **entire** test set rather than a subset.

4.3 Node Classification

The node classification task aims at predicting unknown labels of nodes in a graph. For the transductive setting, the whole graph structure is available and we need to predict the labels of the test set with training labels. Here we use the official split for Cora, Citeseer,

PubMed and Ogbn-Arxiv [22], and split the Sailing with 40%-30%-30% for train, validation and test. For the inductive setting, only nodes that appear in the training set are available during training, and the whole graph structure is available for inference. We split the Flickr and Reddit with 20%-40%-40% for train, validation and test, and use the same split as transductive setting for Sailing.

Transductive Results. Table 2 reports the results of all the methods on the transductive node classification task with uniform and structural missing types. DART outperforms all the baselines across every dataset. Specifically, on the four public datasets with uniformly missing features, DART achieves consistent improvements, surpassing the strongest baseline PCFI by margins ranging from 1.71% to 2.80%. Notably, on the real-world Sailing dataset, our method achieves a significant improvement of 2.43% compared to the runner-up PCFI, demonstrating its robustness in handling natural feature sparsity. In scenarios with structurally missing features, DART maintains its superiority. On the Cora dataset, for instance, it surpasses PCFI by an accuracy margin of 3.14%. Overall, methods utilizing feature imputation generally outperform standard GNNs like GAT and GCN, which suffer performance drops as they assume complete features. An exception is GCNMF, whose performance significantly declines under high missing rates. FP performs relatively well in transductive settings due to its diffusion mechanism, but DART achieves the best results by effectively integrating feature rectification and structural augmentation.

Inductive Results. We further evaluate the performance under the inductive setting, as shown in Table 3. Since ASD-VAE is not designed for inductive tasks and runs out-of-memory on large graphs like Reddit, it is excluded from this comparison. The results indicate that baselines relying solely on simple propagation, such as FP and PCFI, often result in suboptimal performance due to the feature distribution shift between training and inference graphs. Similarly, standard GNNs struggle to generalize to unseen nodes with missing features. In contrast, DART effectively bridges this gap by utilizing masked autoencoders to learn the underlying feature distribution. Consequently, DART achieves substantial gains across all inductive benchmarks. For example, on the Sailing dataset, DART outperforms the second-best method PaGNN by a remarkable margin of 12.71% and PCFI by 14.04%, validating the effectiveness of our proposed distribution rectification mechanism.

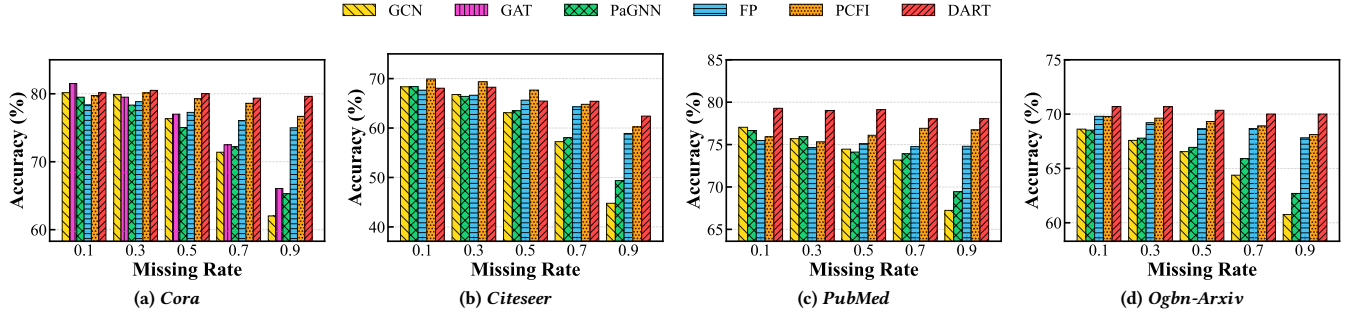
Robustness to Missing Rates. Additionally, we evaluate the performance across varying missing rates (ranging from 10% to 90%) on four datasets: Cora, Citeseer, PubMed, and Ogbn-Arxiv. ASD-VAE and DRI are excluded from large datasets due to out-of-memory issues, and GCNMF is omitted due to its consistently inferior performance. The results, visualized in Figure 4, reveal that most methods not based on feature propagation experience significant performance degradation as the missing rate increases. In contrast, DART consistently achieves superior performance across different missing rates, underscoring its robustness in handling severe feature sparsity for real-world datasets.

4.4 Link Prediction

The link prediction task involves predicting missing or unknown edges within a graph. We conduct experiments on five datasets:

Table 2: Transductive node classification results (Sailing with 80.4% missing rate, others with 90%).

Missing Type	Dataset	GAT	GCN	GCNMF	PaGNN	FP	PCFI	ASD-VAE	DRI	DART
Uniform	Cora	66.06 $\pm$ 1.37	62.02 $\pm$ 0.95	42.94 $\pm$ 2.36	65.30 $\pm$ 1.31	74.98 $\pm$ 0.88	<u>76.68 $\pm$ 1.55</u>	71.94 $\pm$ 0.76	72.88 $\pm$ 0.48	79.48 $\pm$ 0.30
	Citeseer	47.68 $\pm$ 1.30	44.78 $\pm$ 2.09	30.14 $\pm$ 4.68	49.36 $\pm$ 1.78	58.90 $\pm$ 1.00	<u>60.22 $\pm$ 0.80</u>	56.94 $\pm$ 0.94	57.46 $\pm$ 1.04	62.64 $\pm$ 0.74
	PubMed	67.38 $\pm$ 1.24	67.24 $\pm$ 1.84	45.00 $\pm$ 3.04	69.44 $\pm$ 1.01	75.80 $\pm$ 0.62	<u>76.74 $\pm$ 0.47</u>	OOM	70.88 $\pm$ 0.54	78.88 $\pm$ 0.40
	Ogbn-Arxiv	58.74 $\pm$ 0.21	60.76 $\pm$ 0.28	57.36 $\pm$ 0.74	62.49 $\pm$ 0.20	67.84 $\pm$ 0.40	<u>68.27 $\pm$ 0.18</u>	OOM	OOM	69.98 $\pm$ 0.22
Structural	Cora	54.88 $\pm$ 1.43	53.18 $\pm$ 2.19	24.80 $\pm$ 6.72	55.26 $\pm$ 2.31	73.06 $\pm$ 1.86	<u>73.98 $\pm$ 1.61</u>	71.00 $\pm$ 1.01	72.02 $\pm$ 0.56	77.12 $\pm$ 0.82
	Citeseer	40.52 $\pm$ 1.90	37.30 $\pm$ 1.95	20.02 $\pm$ 2.53	40.34 $\pm$ 2.26	55.44 $\pm$ 1.44	<u>59.38 $\pm$ 1.02</u>	54.52 $\pm$ 1.39	56.22 $\pm$ 0.76	60.08 $\pm$ 0.42
	PubMed	63.28 $\pm$ 1.48	62.10 $\pm$ 2.00	40.46 $\pm$ 4.15	64.74 $\pm$ 1.09	73.78 $\pm$ 0.35	<u>75.08 $\pm$ 2.40</u>	OOM	68.96 $\pm$ 0.32	78.02 $\pm$ 0.59
	Ogbn-Arxiv	60.97 $\pm$ 0.33	56.83 $\pm$ 0.43	34.89 $\pm$ 4.36	60.70 $\pm$ 0.42	<u>67.77 $\pm$ 0.15</u>	<u>67.73 $\pm$ 0.28</u>	OOM	OOM	68.24 $\pm$ 0.29
Natural	Sailing	63.07 $\pm$ 1.47	64.45 $\pm$ 0.62	48.66 $\pm$ 2.53	67.47 $\pm$ 1.11	66.91 $\pm$ 0.90	<u>68.45 $\pm$ 0.36</u>	66.71 $\pm$ 1.02	64.62 $\pm$ 0.96	70.88 $\pm$ 0.96

**Figure 4: Performance under different missing rates for transductive learning (best viewed in color).****Table 3: Inductive node classification results.**

Missing Type	Dataset	GAT	GCN	GCNMF	PaGNN	FP	PCFI	DRI	DART
Uniform	Flickr	43.90 $\pm$ 0.94	44.99 $\pm$ 0.33	47.73 $\pm$ 2.71	<u>49.46 $\pm$ 0.12</u>	48.10 $\pm$ 0.77	47.72 $\pm$ 0.47	45.14 $\pm$ 0.28	51.98 $\pm$ 0.08
	Reddit	49.90 $\pm$ 2.50	84.08 $\pm$ 0.85	OOM	<u>86.77 $\pm$ 0.50</u>	86.28 $\pm$ 2.69	OOM	OOM	93.88 $\pm$ 0.21
Structural	Flickr	42.48 $\pm$ 0.09	42.78 $\pm$ 0.07	43.09 $\pm$ 1.01	45.72 $\pm$ 0.29	45.94 $\pm$ 1.59	45.25 $\pm$ 0.68	43.24 $\pm$ 0.37	50.64 $\pm$ 0.20
	Reddit	77.96 $\pm$ 2.68	84.86 $\pm$ 0.97	OOM	<u>90.57 $\pm$ 0.32</u>	89.47 $\pm$ 1.91	OOM	OOM	92.32 $\pm$ 0.38
Natural	Sailing	59.64 $\pm$ 0.97	61.62 $\pm$ 0.60	45.10 $\pm$ 4.84	<u>64.85 $\pm$ 0.56</u>	63.44 $\pm$ 0.98	63.52 $\pm$ 0.21	61.86 $\pm$ 0.41	77.56 $\pm$ 0.24

Table 4: Link prediction results.

	Dataset	GCNMF		GCN		FP		PCFI		DART	
		AUC	AP	AUC	AP	AUC	AP	AUC	AP	AUC	AP
Uniform	Cora	61.09 $\pm$ 1.54	61.73 $\pm$ 2.23	71.13 $\pm$ 2.19	72.86 $\pm$ 2.62	<u>86.65 $\pm$ 0.68</u>	<u>89.21 $\pm$ 0.35</u>	86.16 $\pm$ 0.89	88.91 $\pm$ 0.46	94.62 $\pm$ 0.90	94.28 $\pm$ 0.79
	Citeseer	59.00 $\pm$ 1.72	58.54 $\pm$ 2.09	72.31 $\pm$ 2.44	75.65 $\pm$ 2.27	<u>81.18 $\pm$ 1.17</u>	<u>86.02 $\pm$ 1.30</u>	80.68 $\pm$ 1.38	85.44 $\pm$ 1.01	93.90 $\pm$ 0.96	93.02 $\pm$ 0.87
	PubMed	61.62 $\pm$ 0.51	69.46 $\pm$ 0.67	<u>82.18 $\pm$ 0.77</u>	82.33 $\pm$ 0.45	81.15 $\pm$ 0.19	85.61 $\pm$ 0.19	80.72 $\pm$ 1.02	<u>85.62 $\pm$ 0.56</u>	97.28 $\pm$ 0.18	97.04 $\pm$ 0.17
	Ogbn-Arxiv	84.87 $\pm$ 1.02	87.09 $\pm$ 0.93	90.97 $\pm$ 0.43	91.52 $\pm$ 0.44	94.71 $\pm$ 0.31	95.20 $\pm$ 0.28	<u>95.72 $\pm$ 0.10</u>	<u>96.04 $\pm$ 0.10</u>	98.78 $\pm$ 0.08	98.88 $\pm$ 0.07
Structural	Cora	58.52 $\pm$ 1.19	63.83 $\pm$ 1.49	67.86 $\pm$ 1.65	70.15 $\pm$ 1.55	<u>85.24 $\pm$ 1.55</u>	87.54 $\pm$ 1.26	85.11 $\pm$ 1.27	<u>87.96 $\pm$ 1.02</u>	94.56 $\pm$ 1.20	93.92 $\pm$ 1.61
	Citeseer	56.39 $\pm$ 1.94	63.32 $\pm$ 1.75	62.93 $\pm$ 2.25	66.21 $\pm$ 1.46	80.07 $\pm$ 1.31	84.06 $\pm$ 1.42	<u>81.53 $\pm$ 1.28</u>	<u>85.00 $\pm$ 1.44</u>	93.78 $\pm$ 0.48	93.04 $\pm$ 0.82
	PubMed	63.25 $\pm$ 0.37	72.54 $\pm$ 0.30	<u>86.30 $\pm$ 0.55</u>	84.81 $\pm$ 0.29	79.59 $\pm$ 0.75	84.81 $\pm$ 0.47	82.93 $\pm$ 0.41	<u>86.62 $\pm$ 0.37</u>	97.02 $\pm$ 0.19	96.70 $\pm$ 0.21
	Ogbn-Arxiv	84.97 $\pm$ 1.02	87.26 $\pm$ 0.87	90.97 $\pm$ 0.43	91.53 $\pm$ 0.43	94.69 $\pm$ 0.29	95.17 $\pm$ 0.25	<u>95.71 $\pm$ 0.09</u>	<u>96.03 $\pm$ 0.10</u>	98.66 $\pm$ 0.11	98.68 $\pm$ 0.08
	Sailing	95.78 $\pm$ 0.20	95.25 $\pm$ 0.24	96.03 $\pm$ 0.20	95.21 $\pm$ 0.32	97.89 $\pm$ 0.12	97.67 $\pm$ 0.17	<u>97.97 $\pm$ 0.08</u>	<u>97.80 $\pm$ 0.09</u>	98.52 $\pm$ 0.12	98.38 $\pm$ 0.14

Cora, Citeseer, PubMed, Ogbn-Arxiv, and Sailing. For Cora, Citeseer, PubMed and Ogbn-Arxiv, we randomly sample positive and negative edges for validation (5%) and testing (10%). For the Sailing dataset, we adopt a split of 60%/10%/30% for training, validation, and test. Following previous work [31], we sample the same number

of negative edges as positive edges to train a Graph Autoencoder (GAE) [21] for all methods to predict the link. Performance is measured using the Area Under the Curve (AUC) and Average Precision (AP) metrics.

Table 5: Ablation studies of each mechanism of DART on different datasets.

Variant	Cora	PubMed	Flickr
w/o Rec	78.88 $\pm$ 0.86	77.96 $\pm$ 0.48	50.48 $\pm$ 0.28
w/o GSA	77.10 $\pm$ 1.02	77.98 $\pm$ 0.59	51.93 $\pm$ 0.11
w/o Gaussian Mask	78.16 $\pm$ 1.03	77.92 $\pm$ 0.43	51.64 $\pm$ 0.14
DART (Full)	79.48 $\pm$ 0.30	78.88 $\pm$ 0.40	51.98 $\pm$ 0.08

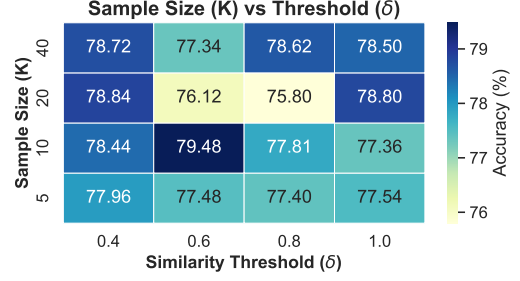
Results. Table 4 summarizes the link prediction performance under uniform and structural missing settings, as well as the naturally missing setting for Sailing. DART consistently achieves superior performance across all datasets and metrics. Specifically, in the uniform missing setting, DART demonstrates substantial improvements over the strongest baselines. For instance, on the PubMed dataset, it outperforms the runner-up by a significant margin of 15.10% in terms of AUC. Across the four public datasets, the AUC improvements range from 3.06% to 15.10%, validating the model’s ability to reconstruct semantically meaningful features that align well with the graph topology. In the structural missing setting, DART maintains its dominance. Notably, on the Citeseer dataset, it achieves an AUC of 93.78%, surpassing the best baseline by 12.25%. This indicates that even when node features are entirely absent, DART can effectively leverage the structural information to recover latent semantics. On the real-world Sailing dataset, which features naturally missing data, DART also achieves the highest performance with an AUC of 98.52% and AP of 98.38%, slightly outperforming the competitive PCFI baseline. This confirms the robustness and practical value of DART in handling complex, real-world missing patterns.

4.5 Ablation Studies

To further investigate the contribution of each mechanism in DART, we conduct ablation studies on three representative datasets: Cora and PubMed (Transductive), and Flickr (Inductive). We compare the full model against three variants: (1) **w/o Rec**: removing the rectification module; (2) **w/o GSA**: removing the Global Structural Augmentation strategy; and (3) **w/o Gaussian Mask**: replacing the Gaussian noise masking with no corruption. The results are summarized in Table 5.

Effect of Rectification. The rectification module is designed to rectify the feature distribution shift. As shown in Table 5, removing this component leads to a consistent performance drop. In the inductive setting (Flickr), the accuracy drops by 1.50%, verifying the necessity of rectification in aligning the propagated features with the true data manifold. Similarly, in the transductive setting, we observe improvements ranging from 0.60% to 0.92%, attributing to the denoising effect that refines the feature representations.

Effect of Global Structural Augmentation (GSA). GSA aims to mitigate topological isolation. The results indicate that GSA is particularly critical for sparse graphs like Cora, where removing it causes a significant performance decline of 2.38%. On PubMed, the drop is moderate (0.90%). In contrast, on the dense Flickr graph, the impact is negligible (0.05%). This aligns with our analysis that dense graphs naturally possess high reachability, whereas GSA serves as

**Figure 5: Sensitive analysis of k and δ .**

a vital bridge for information flow in sparse scenarios, ensuring the robustness of DART across varying graph densities.

Effect of Gaussian Masking. We introduce Gaussian masking to force the autoencoder to learn the gradient field of the data distribution (i.e., the score function). The ablation results show that removing Gaussian masking (w/o Gaussian Mask) leads to performance degradation across all datasets, with drops of 1.32% on Cora and 0.34% on Flickr. This confirms that simple rectification without proper noise injection may lead to learning trivial identity mappings. The Gaussian noise perturbation is essential for learning a robust manifold projector that can effectively pull off-manifold features back to high-density regions.

4.6 Parameter Sensitivity Analysis

We investigate the sensitivity of DART to two key hyperparameters in the GSA module: size K and the similarity threshold δ . We conduct experiments on the Cora dataset with $K \in \{5, 10, 20, 40\}$ and $\delta \in \{0.4, 0.6, 0.8, 1.0\}$.

The results indicate that DART is sensitive to the size K . Specifically, it achieves the highest accuracy when $K = 10$ and $\delta = 0.6$. Performance degrades when K is too small (e.g., $K = 5$) as it limits the semantic receptive field, while an excessively large K (e.g., $K = 40$) introduces noise by connecting to semantically irrelevant nodes. Regarding the threshold δ , moderate values generally yield superior robustness by effectively filtering out low-confidence links. Therefore, a moderate K around 10 and δ larger than 0.6 is recommended to balance information gain and noise control.

4.7 Visualization

To intuitively verify the effectiveness of our proposed feature rectification mechanism, we employ t-SNE [32] to project the high-dimensional feature vectors into a low-dimensional space for visualization. Figure 6 presents a comparative analysis between the raw features and the rectified features (processed by DART). In the transductive setting on the Cora dataset, the raw features exhibit blurred boundaries with loosely scattered nodes, while the feature distribution becomes significantly more discriminative after rectification. This suggests that DART effectively filters out the noise introduced by topological diffusion, providing clearer decision boundaries for downstream classifiers. Furthermore, we visualize the Flickr dataset to assess performance under the more challenging inductive setting. Due to the large scale of Flickr, we randomly sample 3% nodes from the two most representative classes. While the raw features

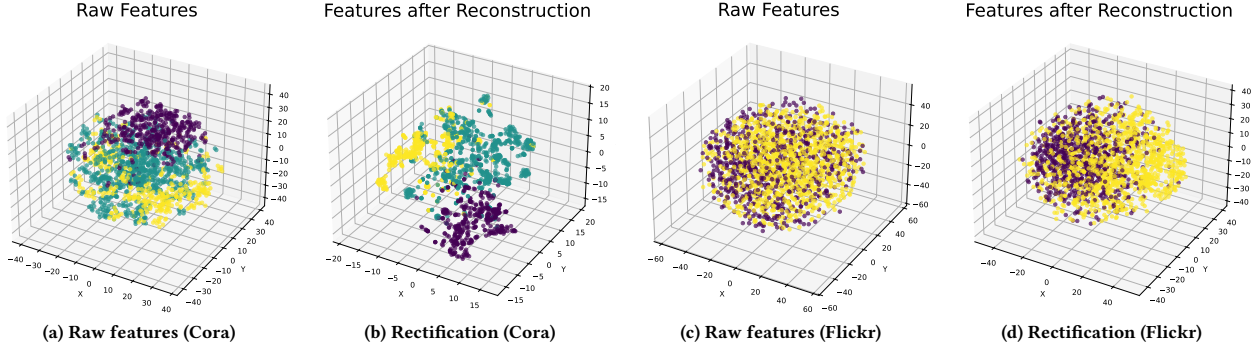


Figure 6: Visualization for transductive (Cora) and inductive (Flickr): raw features and features after rectification.

suffer from severe overlapping due to distribution shift, the visualization reveals that DART successfully mitigates this issue by projecting these drifting features back onto a structured manifold, resulting in a clear separation between the two classes. This visual evidence strongly supports our claim that the MAE module learns a robust, structure-invariant feature distribution, which is crucial for generalizing to unseen graphs.

5 Related Work

Missing node features imputation. Some existing studies [15, 17, 19, 27, 30, 40] explore methods to improve the performance of Graph Neural Networks (GNNs) in handling missing data. SAT [5] introduces a Transformer-like model for feature reconstruction alongside a GNN model to address downstream tasks. GCNMF [30] represents missing data using a Gaussian Mixture Model; however, its performance significantly declines when the proportion of missing features is high. PaGNN [17] proposes two novel partial aggregation (reconstruction) functions specifically designed for incomplete graphs. Feature Propagation (FP) [27], a general approach for managing missing node features in graph machine learning tasks, includes an initial diffusion-based feature reconstruction step followed by a downstream GNN. ADS-VAE [19] applies VAE model to learn the feature imputation, but it needs excessive space consumption and achieves poor performance on complete datasets with multiple connected components. While these methods have shown promising results in transductive graph learning tasks, they are not well-suited for inductive tasks. Also, they are tailored for fully-connected graphs, which makes their performance seriously decline on graphs with many connected components.

Inductive Learning on graphs. Transductive GNNs are limited to making predictions on nodes and edges seen during training. In contrast, inductive GNNs can generalize to previously unseen nodes and edges, making them more versatile for real-world applications. GraphSAGE [8] generates embeddings by sampling and aggregating features from a node’s local neighborhood, enabling the model to infer embeddings for new nodes that were not present during training. GAT [33] assigns different weights to the neighboring nodes during the aggregation process, allowing the model to focus on the most relevant parts of the graph. Some recent studies have developed inductive GNN models for recommender systems [43, 44].

[42] propose a graph sampling approach to build subgraphs for training GNNs on large graph datasets.

Graph Autoencoders. Autoencoders [11] are designed to reconstruct specific inputs based on the given contexts and do not impose any decoding order, unlike autoregressive methods. Graph Autoencoders mostly adopt structural and feature reconstruction as their objectives [26, 34]. Masked Autoencoders (MAEs) [9] are a type of neural network architecture primarily designed for self-supervised learning tasks. GraphMAE [12] leverages the inherent structure of graph data to improve learning and representation by masking and predicting parts of the graph during training.

6 Conclusion

In this paper, we introduced DART, a novel framework that synergizes feature propagation with a distribution-aware rectification mechanism to address missing feature imputation in both transductive and inductive settings. To overcome topological limitations, a Global Structural Augmentation (GSA) strategy is introduced to bridge disjoint components and enhance diffusion reachability. Furthermore, a manifold learner coupled with a test-time rectification scheme is developed to recover semantic diversity and mitigate the feature distribution shift caused by structural changes. To benchmark performance in realistic scenarios, we propose a new maritime dataset Sailing characterized by naturally missing attributes. Extensive experiments on Sailing and six public datasets demonstrate that DART significantly outperforms state-of-the-art methods across node classification and link prediction tasks, highlighting its robustness and generalizability. In the future work, we expect to expand DART to more extensive scenarios (e.g., heterophilic graphs).

A Appendix

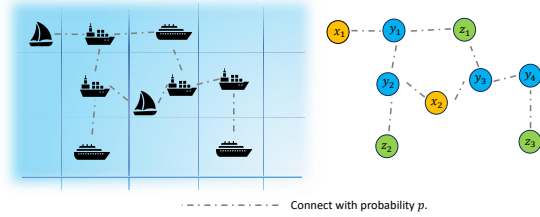
A.1 Feature distribution shift validation

According to Section 2, the core reason behind the feature distribution shift issue is that the distribution of feature imputed by feature propagation strictly rely on the graph structure, make the difference between training and inference. To validate this, we set the function $\mathcal{F}$ as the Kullback-Leibler Divergence and record the Δ_S in Table 6. It could be seen that DART can effectively reduce the feature distribution shift.

Table 6: Δ_S on Flickr, Reddit and Sailing.

Dataset	Flickr	Reddit	Sailing
FP	1.08	0.16	1.96
DART	0.82 ($\downarrow$ 24.07%)	0.10 ($\downarrow$ 37.50%)	1.25 ($\downarrow$ 36.22%)

A.2 Construction of Sailing

**Figure 7: The construction of Sailing dataset.**

Our original Sailing dataset contains information about multiple ships, with each ship represented as a node in a graph. The navigation status of each ship serves as a label, with a total of n labels. Edges between nodes are constructed based on the ships' longitude and latitude coordinates. We divide the entire ocean into several small areas using these coordinates. If two nodes are in the same area or in adjacent areas, we create an undirected edge between them with a 50% probability. The process is shown in Figure 7. The dataset includes both qualitative and quantitative features for each ship. Qualitative features are converted into tensor data using one-hot encoding, while quantitative features are normalized. The Sailing dataset naturally contains missing or invalid data, which we mark as 'NaN'.

References

- [1] Guillaume Alain and Yoshua Bengio. 2014. What regularized auto-encoders learn from the data-generating distribution. *The Journal of Machine Learning Research* 15, 1 (2014), 3563–3593.
- [2] Danish Maritime Authority. 2023. Historical AIS Data. <http://web.ais.dk/aisdata/>.
- [3] Hongyun Cai, Vincent W Zheng, and Kevin Chen-Chuan Chang. 2018. A comprehensive survey of graph embedding: Problems, techniques, and applications. *IEEE transactions on knowledge and data engineering* 30, 9 (2018), 1616–1637.
- [4] Deli Chen, Yankai Lin, Wei Li, Peng Li, Jie Zhou, and Xu Sun. 2020. Measuring and relieving the over-smoothing problem for graph neural networks from the topological view. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 34. 3438–3445.
- [5] Xu Chen, Siheng Chen, Jiangchao Yao, Huangjie Zheng, Ya Zhang, and Ivor W Tsang. 2020. Learning on attribute-missing graphs. *IEEE transactions on pattern analysis and machine intelligence* 44, 2 (2020), 740–757.
- [6] Yuchen Fang, Yanjun Qin, Haiyong Luo, Fang Zhao, Bingbing Xu, Liang Zeng, and Chenxing Wang. 2023. When spatio-temporal meet wavelets: Disentangled traffic forecasting via efficient spectral graph attention networks. In *2023 IEEE 39th International Conference on Data Engineering (ICDE)*. IEEE, 517–529.
- [7] Tom Fawcett. 2006. An introduction to ROC analysis. In *Pattern recognition letters*, Vol. 27. Elsevier, 861–874.
- [8] Will Hamilton, Zhitaoying, and Jure Leskovec. 2017. Inductive representation learning on large graphs. *Advances in neural information processing systems* 30 (2017).
- [9] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. 2022. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 16000–16009.
- [10] Mikael Henaff, Joan Bruna, and Yann LeCun. 2015. Deep convolutional networks on graph-structured data. *arXiv preprint arXiv:1506.05163* (2015).
- [11] Geoffrey E Hinton and Richard Zemel. 1993. Autoencoders, minimum description length and Helmholtz free energy. *Advances in neural information processing systems* 6 (1993).
- [12] Zhenyu Hou, Xiao Liu, Yukuo Cen, Yuxiao Dong, Hongxia Yang, Chunjie Wang, and Jie Tang. 2022. Graphmae: Self-supervised masked graph autoencoders. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 594–604.
- [13] Weihua Hu, Matthias Fey, Marinka Zitnik, Yuxiao Dong, Hongyu Ren, Bowen Liu, Michele Catasta, and Jure Leskovec. 2020. Open graph benchmark: Datasets for machine learning on graphs. *Advances in neural information processing systems* 33 (2020), 22118–22133.
- [14] Mengda Huang, Yang Liu, Xiang Ao, Kuan Li, Jianfeng Chi, Jinghua Feng, Hao Yang, and Qing He. 2022. Auc-oriented graph neural network for fraud detection. In *Proceedings of the ACM web conference 2022*. 1311–1321.
- [15] Cuiying Huo, Di Jin, Yawen Li, Dongxiao He, Yu-Bin Yang, and Lingfei Wu. 2023. T2-gnn: Graph neural networks for graphs with incomplete features and structure via teacher-student distillation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. 4339–4346.
- [16] Aapo Hyvärinen and Peter Dayan. 2005. Estimation of non-normalized statistical models by score matching. *Journal of Machine Learning Research* 6, 4 (2005).
- [17] Bo Jiang and Ziyang Zhang. 2020. Incomplete graph representation and learning via partial graph neural networks. *arXiv preprint arXiv:2003.10130* (2020).
- [18] Bo Jiang, Ziyang Zhang, Doudou Lin, Jin Tang, and Bin Luo. 2019. Semi-supervised learning with graph learning-convolutional networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 11313–11320.
- [19] Xinke Jiang, Zidi Qin, Jiarong Xu, and Xiang Ao. 2024. Incomplete graph learning via attribute-structure decoupled variational auto-encoder. In *Proceedings of the 17th ACM International Conference on Web Search and Data Mining*. 304–312.
- [20] Thomas N Kipf and Max Welling. 2016. Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907* (2016).
- [21] Thomas N Kipf and Max Welling. 2016. Variational Graph Auto-Encoders. In *arXiv e-prints*. <https://arxiv.org/abs/1611.07308>
- [22] Thomas N. Kipf and Max Welling. 2017. Semi-Supervised Classification with Graph Convolutional Networks. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=SJU4ayYgl>
- [23] Xin Li, Xiaowen Ying, and Mooi Choo Chuah. 2019. Grip: Graph-based interaction-aware trajectory prediction. In *2019 IEEE Intelligent Transportation Systems Conference (ITSC)*. IEEE, 3960–3966.
- [24] Dezhi Liu, Richong Zhang, Junfan Chen, Fanshuang Kong, and Jaemin Kim. 2025. Incomplete Graph Learning via Data and Representation-level Interaction. *Knowledge-Based Systems* (2025), 114583.
- [25] Abdulllah Mohamed, Kun Qian, Mohamed Elhoseiny, and Christian Claudel. 2020. Social-stgcn: A social spatio-temporal graph convolutional neural network for human trajectory prediction. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 14424–14432.
- [26] Jiwoong Park, Minsik Lee, Hyung Jin Chang, Kyuewang Lee, and Jin Young Choi. 2019. Symmetric graph convolutional autoencoder for unsupervised graph representation learning. In *Proceedings of the IEEE/CVF international conference on computer vision*. 6519–6528.
- [27] Emanuele Rossi, Henry Kenlay, Maria I Gorinova, Benjamin Paul Chamberlain, Xiaowen Dong, and Michael M Bronstein. 2022. On the unreasonable effectiveness of feature propagation in learning on graphs with missing node features. In *Learning on Graphs Conference*. PMLR, 11–1.
- [28] Prithviraj Sen, Galileo Namata, Mustafa Bilgic, Lise Getoor, Brian Galligher, and Tina Eliassi-Rad. 2008. Collective classification in network data. *AI magazine* 29, 3 (2008), 93–93.
- [29] Yifan Song, Darong Lai, Zhihong Chong, and Zeyuan Pan. 2021. Dynamic Network Embedding by Time-Relaxed Temporal Random Walk. In *Neural Information Processing: 28th International Conference, ICONIP 2021, Sanur, Bali, Indonesia, December 8–12, 2021, Proceedings, Part I 28*. Springer, 426–437.
- [30] Hibiki Taguchi, Xin Liu, and Tsuyoshi Murata. 2021. Graph convolutional networks for graphs containing missing features. *Future Generation Computer Systems* 117 (2021), 155–168.
- [31] Daeho Um, Jiwoong Park, Seulki Park, and Jin young Choi. 2023. Confidence-Based Feature Imputation for Graphs with Partially Known Features. In *The Eleventh International Conference on Learning Representations*. <https://openreview.net/forum?id=YPKBILly-Kt>
- [32] Laurens Van der Maaten and Geoffrey Hinton. 2008. Visualizing data using t-SNE. In *Journal of machine learning research*, Vol. 9.
- [33] Petar Velickovic, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Lio, Yoshua Bengio, et al. 2017. Graph attention networks. *stat* 1050, 20 (2017), 10–48550.
- [34] Chun Wang, Shirui Pan, Guodong Long, Xingquan Zhu, and Jing Jiang. 2017. Mgae: Marginalized graph autoencoder for graph clustering. In *Proceedings of the 2017 ACM on Conference on Information and Knowledge Management*. 889–898.
- [35] Yiwei Wang, Wei Wang, Yuxuan Liang, Yujun Cai, and Bryan Hooi. 2021. Mixup for node and graph classification. In *Proceedings of the Web Conference 2021*. 3663–3674.
- [36] Yiwei Wang, Wei Wang, Yuxuan Liang, Yujun Cai, Juncheng Liu, and Bryan Hooi. 2020. Nodeaug: Semi-supervised node classification with data augmentation. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge*

- Discovery & Data Mining*. 207–217.
- [37] Shunxin Xiao, Shiping Wang, Yuanfei Dai, and Wenzhong Guo. 2022. Graph neural networks in node classification: survey and evaluation. *Machine Vision and Applications* 33, 1 (2022), 4.
- [38] Renchi Yang, Jieming Shi, Keke Huang, and Xiaokui Xiao. 2022. Scalable and effective bipartite network embedding. In *Proceedings of the 2022 International Conference on Management of Data*. 1977–1991.
- [39] Renchi Yang, Jieming Shi, Xiaokui Xiao, Yin Yang, and Sourav S Bhowmick. 2020. Homogeneous network embedding for massive graphs via reweighted personalized pagerank. *Proceedings of the VLDB Endowment* 13, 5 (2020), 670–683.
- [40] Jiaxuan You, Xiaobai Ma, Yi Ding, Mykel J Kochenderfer, and Jure Leskovec. 2020. Handling missing data with graph representation learning. *Advances in Neural Information Processing Systems* 33 (2020), 19075–19087.
- [41] Hanqing Zeng, Hongkuan Zhou, Ajitesh Srivastava, Rajgopal Kannan, and Viktor Prasanna. 2020. GraphSAINT: Graph Sampling Based Inductive Learning Method. In *International Conference on Learning Representations*.
- [42] Hanqing Zeng, Hongkuan Zhou, Ajitesh Srivastava, Rajgopal Kannan, and Viktor Prasanna. 2020. GraphSAINT: Graph Sampling Based Inductive Learning Method. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=BJe8pkHFwS>
- [43] Jiani Zhang, Xingjian Shi, Shenglin Zhao, and Irwin King. 2019. STAR-GCN: stacked and reconstructed graph convolutional networks for recommender systems. In *Proceedings of the 28th International Joint Conference on Artificial Intelligence*. 4264–4270.
- [44] Muhan Zhang and Yixin Chen. 2020. Inductive Matrix Completion Based on Graph Neural Networks. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=ByxxgCEYDS>